

SANDERSON, TX, USA

WMO: 747300

Lat: 30.167N

Lon: 102.417W

Elev: 2838

StdP: 13.25

Time Zone: -6.00 (NAC)

Period: 82-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.8	27.6	6.7	8.6	37.2	10.9	10.7	41.3	22.5	48.4	20.1	47.7	5.2	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	98.7	67.6	96.3	68.2	94.1	68.4	73.7	87.2	72.8	86.7	71.9	86.3	7.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	123.3	79.1	68.9	118.0	78.2	67.8	113.6	77.7	39.3	86.5	38.4	87.2	37.5	86.8	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	15.5	12.7	DB	17.3	104.7	7.8	2.8	11.7	106.8	7.1	108.4	2.7	110.0	-2.9	112.0	
			WB	15.6	75.5	6.4	1.8	11.0	76.7	7.3	77.8	3.7	78.8	-1.0	80.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.9	47.6	52.5	59.0	67.5	75.3	80.6	81.3	81.2	75.2	65.8	55.8	48.6
	DBStd	14.11	7.58	8.69	8.67	7.12	5.77	4.28	3.54	3.42	5.84	7.03	8.49	8.32
	HDD50	398	135	64	26	2	0	0	0	0	0	3	42	126
	HDD65	2059	540	356	220	56	4	0	0	0	7	76	290	510
	CDD50	6212	60	134	305	527	784	917	970	968	757	493	215	82
	CDD65	2398	0	6	34	130	323	467	505	503	315	101	13	1
	CDH74	28008	43	186	569	1796	3529	5401	6079	5820	3118	1170	242	55
	CDH80	13048	5	57	184	741	1695	2740	3087	2858	1256	372	48	5
Wind	WSAvg	6.0	5.8	6.2	7.1	7.0	6.5	6.2	5.9	5.7	5.5	5.2	5.4	5.5
Precipitation	PrecAvg	14.1	0.5	0.7	0.8	0.7	1.5	1.9	2.1	1.7	1.7	1.4	0.7	0.4
	PrecMax	19.7	1.7	2.1	3.4	1.9	5.4	5.2	10.7	4.7	4.7	5.1	3.7	2.0
	PrecMin	8.7	0.0	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.1	0.0	0.0	0.0
	PrecStd	3.4	0.6	0.5	1.0	0.6	1.5	1.6	2.5	1.3	1.5	1.3	0.9	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.0	88.1	89.9	95.9	100.5	103.8	102.2	98.1	96.0	91.4	84.9	80.1
		MCWB	52.7	56.9	58.2	60.8	65.4	69.0	70.0	69.1	68.8	62.6	59.1	54.5
	2%	DB	73.8	79.3	85.4	91.5	97.1	98.3	97.7	95.7	92.7	87.8	79.8	73.8
		MCWB	51.7	54.7	57.0	59.9	65.1	68.4	69.5	69.7	68.1	63.6	57.8	53.3
	5%	DB	69.1	74.1	80.5	88.2	93.1	95.2	94.8	93.9	89.7	84.2	76.5	68.5
		MCWB	49.7	52.3	56.2	59.2	65.1	69.6	69.2	69.8	67.7	63.8	57.6	52.1
	10%	DB	64.0	69.1	75.8	84.2	89.3	92.4	92.3	91.9	87.1	80.7	72.1	64.1
		MCWB	48.2	51.6	55.8	59.2	65.0	69.8	68.7	69.6	67.7	62.8	56.2	51.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.2	59.9	64.7	68.3	73.6	76.0	74.3	73.7	73.2	71.3	65.4	61.9
		MCDB	64.7	69.3	74.3	80.9	87.3	89.2	89.5	89.1	85.8	80.6	72.1	67.5
	2%	WB	54.1	58.0	62.4	66.4	71.6	74.3	73.0	72.6	71.8	69.0	63.6	58.4
		MCDB	66.8	70.3	73.6	78.6	85.6	86.7	88.1	88.5	84.5	78.1	71.0	63.9
	5%	WB	52.0	56.2	60.5	64.8	70.3	73.1	72.0	71.7	70.8	67.5	61.3	55.8
		MCDB	64.9	68.4	71.5	76.6	83.8	85.6	87.3	87.9	83.4	77.2	70.3	65.3
	10%	WB	50.0	54.1	58.4	63.2	69.0	71.9	71.0	70.9	69.8	65.5	59.1	52.5
		MCDB	62.9	66.4	71.4	74.9	81.8	85.4	86.0	86.9	82.2	76.4	69.4	62.7
Mean Daily Temperature Range	5% DB	MDBR	25.8	25.7	26.7	27.8	24.1	22.0	21.8	20.5	21.8	24.7	25.9	24.5
		MCDBR	37.3	35.9	34.2	36.3	31.4	26.6	25.3	23.3	24.8	30.6	30.9	33.6
		MCWBR	19.3	17.8	15.5	15.6	11.5	8.7	8.3	7.8	9.0	13.5	15.0	18.2
	5% WB	MCDBR	30.6	27.3	25.3	25.5	23.0	21.3	20.8	20.3	19.8	21.6	23.2	25.3
		MCWBR	16.9	14.5	13.4	12.2	9.8	7.8	7.2	7.0	7.6	10.4	13.0	15.8
Clear-Sky Solar Irradiance	taub		0.265	0.273	0.289	0.324	0.350	0.371	0.387	0.364	0.353	0.307	0.285	0.266
	taud		2.633	2.600	2.550	2.443	2.411	2.419	2.408	2.496	2.457	2.598	2.582	2.620
	Ebn at Noon		314	319	318	307	297	289	284	291	292	302	302	308
	Edh at Noon		24	28	31	36	38	37	37	34	34	28	26	24
All-Sky Solar Radiation	RadAvg		1148	1430	1813	2138	2262	2344	2226	2089	1768	1487	1211	1050
	RadStd		115	148	106	142	134	110	139	114	118	157	125	92

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166

Nomenclature: See separate page